

Supplementary Appendix: Calibrated Counterfactual Evaluation of Dispatch Policies under Cascading Divergence

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1 Supplementary Tables and Figures

1.1 DDQN Training Summary

Table S1. DDQN training summary (hours 0–48). Full epoch-by-epoch logs available in the supplementary material.

Metric	DDQN
Training epochs	20
Final loss (MSE)	0.077
Final mean Q-value	17.88
Final mean reward	0.229
Final acceptance rate	88.1%
Best lost orders (epoch)	1,779 (Ep. 3)
Final lost orders	2,052
Final late deliveries	22,986

1.2 Training Configuration

Table S2. Hyperparameters for each agent architecture.

Parameter	BC	DDQN
Architecture	DeepMLPBC	QNet (Double DQN)
Hidden layers	[512, 256, 256, 128]	[256, 256, 128]
Activation	GELU	ReLU
Normalisation	LayerNorm	LayerNorm
Dropout	0.0	0.1
Optimiser	AdamW	Adam
Learning rate	10^{-3}	10^{-3}
Batch size	2,048	1,024
Epochs	10	20
Discount (γ)	–	0.99
Target update (τ)	–	0.01
Replay buffer	–	500K (prioritised, $\alpha=0.6$)
Exploration	–	ϵ : 1.0 $\rightarrow$ 0.01

1.3 Lost-Order Feature Profile

Table S3. Top features distinguishing lost from delivered orders (z-score of lost mean relative to delivered distribution).

Rank	Feature	Lost	Delivered	Dir.
1	capacity	0.69	1.16	Lower
2	time_since_accept	3,059s	1,652s	Higher
3	accepts_so_far	10.6	13.7	Lower
4	active_seconds	6,606s	8,635s	Lower
5	late_ratio	0.087	0.126	Lower

1.4 Distance and Reassignment Diagnostics

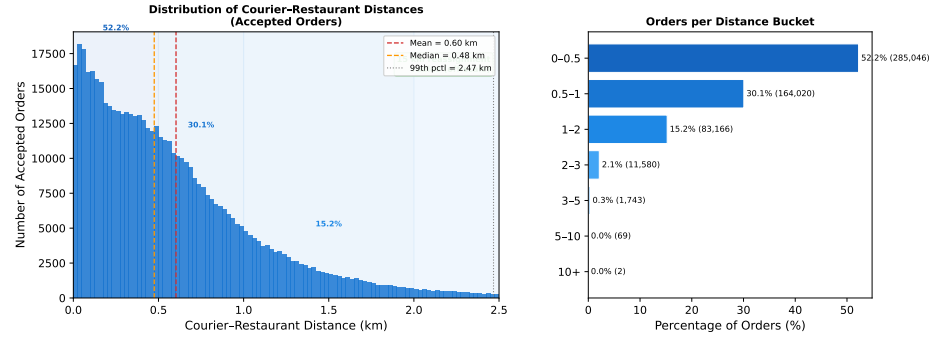


Fig. S1. Left: histogram of haversine distances between couriers and restaurants for accepted orders (cropped at the 99th percentile). Right: percentage of orders in each distance bucket. The vast majority of assignments fall within 2.5 km; the 3 km reassignment radius comfortably covers the entire operational range.

1.5 Dispatch Attempts per Order

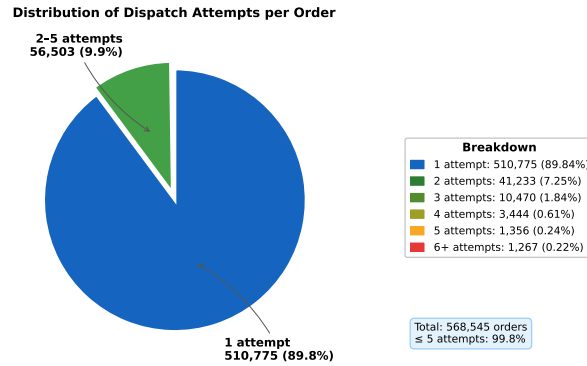


Fig. S2. Distribution of dispatch attempts (waybills) per unique order. 99.8% of orders are resolved within five attempts, justifying the five-cycle reassignment limit in the simulation.